INCH - POUND

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MILITARY SPECIFICATION

GRIP, CABLE

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for strand mesh-type grips used for gripping, pulling and supporting wire, cable and hose.

1.2 Classification. Grips shall be of the following types and classes, as specified (see 6.2):

Type I - Support - riser
  Class 2 - Standard
  Class 3 - Heavy duty
Type II - Support conduit - riser
  Class 1 - Light duty
Type III - End pull
  Class 1 - Light duty
  Class 2 - Standard
  Class 3 - Heavy duty
  Class 4 - Extra heavy duty
  Class 5 - End pull (shipboard)
  Class 6 - Light duty - non conductive
Type IV - Slack pull
  Class 2 - Standard
  Class 3 - Heavy duty

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5120

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standard, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-601	-	Boxes, Wood, Cleated Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
QQ-S-763	-	Steel Bars, Wire, Shapes, and Forgings, Corrosion Resisting

MILITARY

MIL-P-116	-	Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual - Wire and Rods, Carbon Steel - Drawn Galvanized Rope Wire

(Application for copies should be addressed to the American Iron and Steel Institute, 1133 15th Street, NW, Suite 300, Washington, DC 20005.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

A313	-	Standard Specification for Chromium - Nickel Stainless and Heat-Resisting Steel Spring Wire
A342	-	Standard Test Methods for Permeability of Feebly Magnetic Materials
A580	-	Standard Specification for Stainless and Heat-Resisting Steel Wire
B33	-	Tinned Soft or Annealed Copper Wire for Electrical Purposes
B105	-	Hard Drawn Copper Alloy Wires for Electrical Conductors

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified by the procurement document (see 6.2), a first article shall be supplied by the contractor for first article testing under Government surveillance. The first article test shall verify the contractor's capability to produce an item that complies with all the requirements of this specification. The first article may, as specified by the procurement document, be a pre-production sample or an item selected at random by an authorized Government representative from the first month's production. The sample shall be representative of the production quantity, utilizing the same materials, processes, fabrication techniques, assembly techniques, production lines, tools, equipment, and facilities as are used on the production items, and shall be required to meet all of the requirements of this specification and associated engineering drawings, specifications and lists.

3.2 Material. All materials used in the fabrication of cable grips shall be of a quality necessary to produce cable grips meeting the requirements specified herein. Cable grips shall be corrosion resisting steel, drawn galvanized (extra improved plow) steel, tin coated bronze and Aramid fibre, jacketed as specified (see 6.2).

3.2.1 Corrosion resistant steel wire. Corrosion resistant steel wire shall be in accordance with ASTM A313 and ASTM A580. Corrosion resistant steel for type III grip components shall be in accordance with the requirements of QQ-S-763, ASTM A313 and ASTM A580. Drawn galvanized steel shall be in accordance with the requirements of AISI drawn galvanized rope wire and shall be zinc coated (galvanized) or otherwise treated to minimize

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corrosion. Tin coated bronze shall have the wire in accordance with the requirements of ASTM B105 and the tin coating in accordance with the requirements of ASTM B33. The jacketed Aramid fibre shall have the fibres stranded and shall be so coated or jacketed to prevent abrasion and weathering. The jacketed Aramid fibre shall be nonconductive.

3.3 Design.

3.3.1 Body. All cable grips shall be designed so that eccentric loads will not be induced by the method of support or pull. The body (mesh) weave shall consist of a number of single strands and shall be designed to permit the pressure to be distributed evenly over the gripping area. The length of the body (mesh) shall be sufficient to provide for an instant and constant hold. The ends shall be connected in such a way that no sharp edges or ends endanger the workman or adjacent materials. No heat or solder shall be used when the grip is assembled except in non-load bearing areas. If the grip is designed for more than one cable size, it shall perform satisfactorily on all sizes within the design range. The grips shall be made as light as possible, consistent with the grips capacity and approximate breaking strength requirements specified herein. All grips shall be of such design that will prevent breakage or distortion from static or tensional loads specified in 4.5.1. Body length (at nominal diameter) variations may be ± 20 percent of the dimension indicated for the type and class specified herein.

3.3.2 Node. The design of the node shall provide a uniform transition between the body and eye and shall prevent choking of the wire, cable or wire rope being held.

3.3.3 Eye. The eye design shall provide for a termination for supporting or pulling that will permit satisfactory performance of the grip. Eye length variations may be ± 20 percent of the dimension indicated for the type and class specified herein.

3.3.4 Approximate Breaking Strength. Approximate breaking strength shall represent the average strength of new unused grips and shall be not less than 80 percent of the indicated value of the type and class specified herein when tested in accordance with 4.5.3.

3.3.4.1 Safety factor. A safety factor to ensure performance shall be used on the approximate breaking strength to establish a working load. For normal conditions with a new grip, a safety factor of 5 for pulling grips and 10 for support grips may be used. When the application conditions are not well defined or when there is a higher risk of injury to persons or property involved, a greater safety factor should be used. All conditions of the application, including the condition of the grip (whether new or used), are to be evaluated before a safety factor is determined. Ultimately, the safety factor used is the responsibility of the user and his evaluation of the tensions involved.

3.4 Construction. Cable grips shall be free of any characteristics which will prevent the cable grip from conforming to the examination and tests specified in 4.4.3 and 4.5.

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3.5 Surface finish. Surfaces shall be smooth and finished according to the manufacturer's standard practice.

3.6 Identification. Each cable grip shall be marked permanently and legibly when space permits with the manufacturer's part number and the range of sizes for which the grip is designed. The limiting size shall be given, such as 1.50 to 2.00.

3.7 Magnetic permeability. When specified (see 6.2) cable grips of austenitic chrome nickel (corrosion-resisting) steel shall have a magnetic permeability of 2.0 or less after fabrication.

3.8 Types and classes. Grips shall be of the types, classes and sizes specified in 3.8.1 through 3.8.11 as specified (see 1.2 and 6.2). When specified (see 6.2) corrosion-resisting steel grips shall be furnished for type I, class 2 and class 3, and type II, class 1.

3.8.1 Type I, class 2. Type I, class 2 support-riser (standard-duty) grips shall be furnished single weave with single "U" eye, double eye, offset eye, universal bale or hook ending arms (see figure 7). Eye configuration shall be as specified (see 6.2). Unless otherwise specified in the contract or order, grip mesh shall be made of tin coated bronze strand.

3.8.1.1 Grips. Grips shall be closed or split, with stranded lace or metal rod (see figure 1), as specified (see 6.2).

3.8.1.2 Support grips. Capacity, eye length, body length and approximate breaking strength shall be as shown in table I, and as specified (see 6.2).

3.8.2 Type I, class 3. Type I, class 3 support-risers (heavy duty) grips shall be furnished double weave with a single "U" eye or double eye (see 6.5, figure 7) and shall be closed or split with stranded lace, as specified (see 6.2). Unless otherwise specified in the contract or order, grip mesh shall be made of tin coated bronze strand.

3.8.2.1 Support grips. Capacity, eye length, body length and approximate breaking strength shall be as shown in table II, and as specified (see 6.2).

3.8.3 Type II, class 1. Type II, class 1 support conduit-riser (light duty) grips shall be furnished with ring support (see 6.5, figure 2) or hook ends (see 6.5, figure 7), as specified (see 6.2). Single weave grips shall be closed or split with rod closing as specified (see 6.2). Double weave grips shall be closed or split with stranded lace, as specified (see 6.2). Unless otherwise specified in the contract or order, grip mesh shall be made of tin coated bronze strand.

3.8.3.1 Support grips. Capacity, conduit size, body length and approximate breaking strength shall be as shown in table III and as specified (see 6.2).

3.8.4 Type III, class 1. Type III, class 1 end pull (light duty) grips for light pull electrical construction or light wiring and rewiring shall be

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furnished in single weave with a flexible eye (see 6.5, figure 3). All grips shall have shoulders to protect the ends of the pulled cable. Unless otherwise specified in the contract or order mesh shall be made of galvanized steel strand.

3.8.4.1 Support grips. Capacity, eye length, body length and approximate breaking strength shall be as shown in table IV and as specified (see 6.2).

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TABLE I. Support grips (type I, class 2; standard).

Range Code	Capacity (Diameter Range)	Eye Length 1/					Body Length 1/	Breaking Strength 2/				
		Single	Double	Offset	Univ. Bale	Hook Ends		Single	Double	Univ. Bale	Offset	Hook Ends
1	0.50 - 0.62	7	4	4	18	10	9		5		5	4.5
	0.63 - 0.74	8	4	4	18	10	9		8		8	7
	0.75 - 0.99	8	4	4	18	10	12		10		10	10
	1.00 - 1.24	9	5	5	18	10	13		16		15	14
	1.25 - 1.49	10	5	5	18	10	15		16		15	14
	1.50 - 1.74	12	5	5	18	10	16		16		15	14
	1.75 - 1.99	14	6	6	18	10	18		22		20	16
	2.00 - 2.49	16	6	6	18	10	20		33		31	22
	2.50 - 2.99	18	6	6	18	10	22		33		31	22
	3.00 - 3.49	21	8.5	8.5	18	10	24		49		38	32
	3.50 - 3.99	24	8.5	8.5	18	10	26		49		38	32

1/ (In inches)
2/ (Approximate; in 100 pounds)

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3.8.5 Type III, class 2. Type III, class 2 end pull (standard duty) grips for power and communication cables and service lines into buildings shall be furnished in double weave with forged eye, forged eye with swivel, rotating eye, or flexible eye (see 6.5, figures 3 and 4), as specified (see 6.2). All grips shall have shoulders to protect the ends of the pulled cable. Unless otherwise specified in the contract or order grip mesh shall be made of galvanized steel strand.

3.8.5.1 Support grips. Capacity, flexible and forged eye, eye and body length and approximate breaking strength shall be as shown in table V and as specified (see 6.2).

3.8.6 Type III, class 3. Type III, class 3 end pull (heavy duty) grips for normal overhead transmission and distribution line stringing and for bare or insulated conductors shall be furnished in multiweave (single, double or triple) with forged eye, forged eye and swivel, or flexible eye (see 6.5, figures 3 and 4), as specified (see 6.2). All grips shall have shoulders to protect the ends of the cable. Grip mesh shall be made of galvanized steel strand.

3.8.6.1 Support grips. Capacity, forged eye, flexible eye, body length and approximate breaking strength shall be as shown in table VI and as specified (see 6.2).

Table II. Support grips (type I, class 3).

Range Code	Capacity (diameter range) 1/	Single Eye 1/	Double Eye 1/	Body Length 1/	Breaking Strength 2/
01	0.75 - 0.99	10	10	25	28
02	1.00 - 1.24	10	10	28	43
03	1.25 - 1.49	10	10	30	43
04	1.50 - 1.99	10	10	34	43
05	2.00 - 2.49	--	10	36	81
06	2.50 - 2.99	--	10	38	81
07	3.00 - 3.49	--	10	40	101
08	3.50 - 3.99	--	10	44	121
09	4.00 - 4.49	--	10	46	121
10	4.50 - 4.99	--	10	60	181
11	5.00 - 5.99	--	10	72	141

1/ (In inches)
2/ (Approximate; in 100 pounds)

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Table III. Support conduit-riser (type II, class 1).

Range Code	Capacity (diameter range) 1/	Conduit Size 1/, 2/	Body Length 1/	Breaking Strength 3/	
				Single Weave	Double Weave
01	0.75 - 1.00	1.25 - 6.00	13	6	9
02	1.00 - 1.25	1.50 - 6.00	13	9	12
03	1.25 - 1.50	2.00 - 6.00	14	9	12
04	1.50 - 1.75	2.50 - 6.00	15	9	12
05	1.75 - 2.00	2.50 - 6.00	16	12	15
06	2.00 - 2.50	3.00 - 6.00	18	18	24
07	2.50 - 3.00	3.50 - 6.00	19	18	24
08	3.00 - 3.50	4.00 - 6.00	21	23	30
09	3.50 - 4.00	4.50 - 6.00	22	23	30

1/ (In inches)
2/ (Conduit support rings shall be suitable for use with the following standard iron pipe size (IPS) conduit sizes, as specified (see 6.2): 1.25, 1.50, 2.0, 2.50, 3.0, 3.50, 4.0, 4.50, 5.0 and 6.0 inches.)
3/ (Approximate; in 100 pounds)

Table IV. End pull grip (type III, class 1).

Range Code	Capacity (diameter range) 1/	Eye Length 1/	Body Length 1/	Breaking Strength 2/
01	0.50 - 0.61	5	16	28
02	0.62 - 0.74	5	16	28
03	0.75 - 0.99	6	20	40
04	1.00 - 1.24	7	20	68
05	1.25 - 1.49	7	21	68
06	1.50 - 1.99	8	23	68
07	2.00 - 2.49	9	25	85
08	2.50 - 2.99	9	27	106
09	3.00 - 3.49	10	30	147
10	3.50 - 3.99	10	32	147

1/ (In inches)
2/ (Approximate; in 100 pounds)

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Table V. End pull grips (type II, class 2).

Range Code	Capacity (diameter range) 1/	Flexible Eye			Forged Eye			
		Eye Length 1/	Body Length 1/	Breaking Strength 2/	Eye 1/ Dia.	Eye 1/ Length	Body 1/ Length	Breaking Strength 2/
01	0.50-0.61	--	--	--	.875	5	16	56
02	0.62-0.74	--	--	--	.875	5	16	68
03	0.75-0.99	9	36	68	1.00	6	32	96
04	1.00-1.49	9	36	96	1.375	7	33	164
05	1.50-1.99	11	36	164	1.375	7	34	164
06	2.00-2.49	12	36	185	1.625	9	36	272
07	2.50-2.99	12	36	245	1.875	10	38	330
08	3.00-3.49	14	36	245	1.875	10	39	410
09	3.50-3.99	14	40	310	1.875	10	41	480

1/ (In inches)
2/ (Approximate; in 100 pounds)

Table VI. End pull grips (type III, class 3).

Range Code	Capacity (diameter range) 1/	Forged Eye 1/		Flexible Eye Length 1/	Body Length	Breaking Strength 2/
		Dia.	Length			
01	0.25-0.29	.875	5	9	26	68
02	0.50-0.74	1.00	6	9	32	100
03	0.75-0.99	1.00	6	11	41	144
04	1.00-1.24	1.375	8	12	52	246
05	1.25-1.49	1.625	8	12	56	306
06	1.50-1.74	1.875	9	12	60	306
07	1.75-2.24	1.875	10	18	70	480

1/ (In inches)
2/ (Approximate; in 100 pounds)

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3.8.7 Type III, Class 4. Type III, class 4 end pull (extra heavy duty) grips for extra high strength overhead transmission line stringing, for bare and insulated conductors and for synthetic rope and wire rope shall be furnished in double weave with a flexible eye (see 6.5, figure 3). All grips shall have shoulders to protect the end of the cable. Grip mesh shall be made of galvanized steel strand.

3.8.7.1 Support grips. Grip capacity, flexible eye and body length and approximate breaking strength shall be as shown in table VII and as specified (see 6.2).

Table VII. End pull grips (type III, class 4).

Range Code	Capacity (diameter range) 1/		Flexible Eye Length 1/	Body Length 1/	Breaking Strength 2/
	Conductor	Rope			
01	0.19-0.37	0.25-0.65	10	24	65
02	0.38-0.62	0.50-0.90	12	36	140
03	0.63-0.87	0.75-1.10	13	48	200
04	0.88-1.12	1.00-1.50	15	60	296
05	1.13-1.37	1.25-1.70	18	76	468
06	1.38-1.90	1.50-2.10	24	89	665

1/ (In inches)
2/ (Approximate; in 100 pounds)

3.8.8 Type III, class 5. Type III, class 5 end pull (shipboard) grips for shipboard or towing cable and for alternate, constant and intermittent strain and wet and dry conditions shall be furnished double weave, closed or split with stranded lace, as specified (see 6.2) and with a single or double eye, as specified (see 6.2). Grips shall be made of corrosion resisting steel with a magnetic permeability of 2.0 or less after fabrication.

3.8.8.1 Support grips. Capacity, eye and body length and approximate breaking strength shall be as shown in table VIII and as specified (see 6.2).

3.8.9 Type III, class 6. Type III, class 6 end pull (light duty-non-conductive) grips for light overhead distribution line stringing of insulated conductors in close proximity to energized lines or hardware shall be furnished in double weave with a flexible eye (see 6.5, figure 3). All grips shall have shoulders or a sleeve to protect the ends of the cable. The grip mesh shall be made of jacketed aramid fibre.

3.8.9.1 Support grips. Capacity, eye length, body length and approximate breaking strength shall be as shown in table IX and as specified (see 6.2).

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Table VIII. End pull grips (type III, class 5).

Range Code	Capacity (diameter range) 1/	Eye Length 1/	Body Length 1/	Breaking Strength 2/
01	1.25-1.50	15	50	70
02	2.00-2.25	18	59	130
03	2.75-3.00	18	59	130
04	5.00-5.50 3/	18	70	200
05	7.00-8.00 3/	18	81	300

1/ (In inches)
2/ (Approximate; in 100 pounds)
3/ (5-inch through 8-inch capacity grips shall be equipped with double eye only)

Table IX. End pull grips (type III, class 6).

Range Code	Capacity (diameter range) 1/	Eye Length 1/	Body Length 1/	Breaking Strength 2/
01	0.50-0.62	5.5	24	28
02	0.63-0.74	5.5	26	35
03	0.75-0.99	6.0	31	42
04	1.00-1.24	6.5	36	42
05	1.25-1.49	6.7	42	42

1/ (In inches)
2/ (Approximate; in 100 pounds)

3.8.10 Type IV, class 2. Type IV, class 2 slackpull (standard duty) grips for removing cable from conduit and for pulling slack for final placing of new cable shall be furnished with offset flexible eye or ring (see 6.5, figures 5 and 6) as specified (see 6.2). Offset flexible eye grips shall be double weave, closed or split mesh with stranded lace or single weave, split mesh with metal rod, as specified (see 6.2). Ring grips shall be triple weave, closed mesh or double weave, split mesh with rawhide lace, as specified (see 6.2). Grip mesh shall be made of galvanized steel strand.

3.8.10.1 Support grips. Capacity, eye and body length and approximate breaking strength shall be as shown in table X and as specified (see 6.2).

3.8.11 Type IV, class 3. Type IV, class 3 slack, pull (heavy duty) grips for removing underground cable and for pulling slack for final placing of new cable shall be furnished double weave, closed or split mesh, with rawhide lace and with offset flexible eye (see 6.5, figure 5) as specified (see 6.2). Grip mesh shall be made of galvanized steel strand.

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3.8.11.1 Support grips. Capacity, eye and body length and approximate breaking strength shall be as shown in table XI and as specified (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. Unless otherwise specified (see 6.2), the first article required by 3.1 shall consist of one of each type and class of cable grip submitted for first article inspection. The sample submitted shall be representative of production items. Unless otherwise specified, the first article shall undergo examination and tests specified in 4.4.3 and 4.5. The examinations and tests shall be conducted in the sequence specified therein and shall verify compliance with all requirements specified in Sections 3 and 5 herein. Failure to comply with any requirement shall constitute cause for disapproval of the first article. Unless otherwise specified, acceptance of any production items shall be withheld pending approval of the first article.

4.4 Quality conformance inspection. The quality conformance inspections shall be performed as specified in tables XII and XIII for verification of the corresponding requirements of sections 3 and 5 indicated therein. The production lots shall comply with all of the requirements specified herein. Failure of any item in the lot constitutes rejection of the lot. Such items will be removed from the lot and replaced with items that have passed the quality conformance inspections specified herein.

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Table X. Slack pull grips (type IV, class 2).

Range Code	Capacity (diameter range) 1/	Offset Flexible Eye					Ring		Breaking Strength 2/	
		Eye Length 1/	Body Length 1/	Breaking Strength 2/			Eye Length 1/	Body Length 1/		
				Closed	Split Lace	Split Rod			Closed	Split Lace
11	0.75-0.99	7	11	26	25	22	11	13	50	40
12	1.00-1.24	8	13.5	40	35	34	11	14	50	40
13	1.25-1.49	8	15	54	40	45	11	15	50	40
14	1.50-1.74	9	16	66	40	58	11	16	50	40
15	1.75-1.99	10	17	100	40	76	13	17	50	50
16	2.00-2.49	10	19	110	40	90	13	17	50	50
17	2.50-2.99	10	20	110	40	110	13	18	50	50
18	3.00-3.49	12	21	145	40	120	15	19	100	50
19	3.50-3.99	12	23	145	40	120	15	20	100	50

1/ (In inches)
2/ (Approximate, in 100 pounds)

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Table XI. Slack end pull grips (type IV, class 3).

Range Code	Capacity (diameter range) 1/	Eye Length 1/	Body Length 1/	Breaking Strength 2/	
				Closed	Split
01	0.75-0.99	7	20	26	25
02	1.00-1.24	8	20	54	35
03	1.25-1.49	8	23	54	40
04	1.50-1.99	9	25	66	40
05	2.00-2.49	10	26	110	40
06	2.50-2.99	10	29	110	40
07	3.00-3.49	12	32	160	40
08	3.50-3.99	12	35	160	40

1/ (In inches)
2/ (Approximate; in 100 pounds)

4.4.1 Lot formation. An inspection lot shall consist of those like items produced at the same place using the same batches of materials, lots of components, process runs, fabrication techniques, assembly techniques, tools, equipment, and facilities, but shall not exceed one month's production. Upon submittal to the procuring activity for acceptance, each lot shall be accompanied by in-process inspection records and inspection results as required by the procuring activity inspector and this specification.

4.4.2 Sampling. Sampling for quality conformance inspection shall be in accordance with table XII. All characteristics are classified major and any defect shall constitute failure of the represented lot.

4.4.3 Examination. Each cable grip selected for quality conformance inspection shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with table XIII. Nonconforming cable grips shall not be offered for delivery and if there are nonconforming cable grips in any one sample, the lot represented shall be rejected.

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Table XII. Sampling for quality conformance.

Lot Size	Sample Size
2 - 5	100%
6 - 50	5
51 - 90	7
91 - 150	11
151 - 280	13
281 - 500	16
501 - 1200	19
1201 - 3200	23
3201 - 10,000	29
10,001 - 35,000	35
35,001 and over	40

Table XIII. Quality conformance inspection.

Characteristic	Requirement	Inspection	Inspection Method	Inspection Level
Body Length	3.3.1 per class type and diarange	4.4.3	Measure	Table XII
Eye Length	3.3.3 per class type and diarange	4.4.3	Measure	Table XII
Magnetic Permeability	3.7	4.5.4	ASTM A342	Table XII
Tensile Strength	3.3.1	4.5.1	Test	Table XII

4.5 Tests.

4.5.1 Tensile test. The tensile tests specified in 4.5.2 and 4.5.3 shall be performed with new grips on metal rod having a nominal maximum diameter for the grip size tested. The inherent hardness of the metal rod, in addition to its low frictional characteristics, will not in all instances permit gripping of sufficient tightness to overcome slippage during the performance of a test. To readily conform to test requirements and provide for completion of the tests, a clamping or banding device is to be applied to the open end of the grip body (mesh) with sufficient force to prevent slippage.

4.5.2 Loading. The test shall be performed with a tensile load equal to half of the manufacturers stated approximate breaking strength. After 2 minutes of load, the load shall be released, the grip shall be removed and examined for frayed or broken strands or evidence of permanent distortion.

4.5.2.1 Testing type III, class 5 grips. The test for type III, class 5 grips shall have the eye(s) at an angle of 15 degrees off the centerline and the load applied shall be held for 5 minutes at the load strength specified.

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4.5.3 Approximate breaking strength test. After subjection to the performance test, specified in 4.5.2 the cable grip assembly shall be tensioned until breakage occurs. Approximate breaking strengths shall be as specified in the tables herein as applicable for the class and type cable grip tested and in accordance with 3.3.4.

4.5.4 Magnetic permeability. Magnetic permeability testing shall be performed in accordance with ASTM A 342, to determine compliance to 3.7.

4.6 Inspection of delivery preparation. Packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PACKAGING

5.1 Preservation. Packaging, packing, unitization, and marking shall be level A, B, or Commercial (see 6.2).

5.1.1 Level A. The Grip shall be packaged in accordance with MIL-P-116, Method III, utilizing containers conforming to PPP-B-636, Class weather resistant or PPP-B-566, Variety 2.

5.1.2 Level B. Same as level A.

5.1.3 Commercial. Preservation and packaging shall be in accordance with MIL-STD-1190.

5.2 Packing.

5.2.1 Level A. The packaged Grips shall be packed in containers conforming to PPP-B-601, overseas type or PPP-B-621, Class 2, style 4.

5.2.2 Level B. Same as level A.

5.2.3 Commercial. The preserved Grips shall be packed in accordance with the requirements of MIL-STD-1190.

5.3 Marking.

5.3.1 Levels A or B. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with the requirements of MIL-STD-1190.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.1.1 Type I, class 2 grips. These grips are used to absorb flexure and position variations and to support light cable weight.

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6.1.2 Type I, class 3 grips. These grips have the same application as type I, class 2, except for heavier cable weight.

6.1.3 Type II, class 1 grips. These grips are used as a method of support for wire and cable, in vertical ducts and conduit.

6.1.4 Type III, classes 1, 2, 3, 4 and 6 grips. These grips are used to pull cable or synthetic fiber rope through conduits or overhead with various degrees of torque for light to extra heavy loading.

6.1.5 Type III, class 5 grips. These grips are used for shipboard use including towing of cable where cables are subject to alternate, constant and intermittent strain and wet or dry conditions.

6.1.6 Type IV, classes 2 and 3 grips. These grips are used for pulling slack or removing cable in conduit with standard and heavy loading.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
c. Type and class required (see 3.7, 3.8 and 6.4)
d. Capacity (diameter range) required (see 3.8.1.1, 3.8.2.1, 3.8.3.1, 3.8.4.1, 3.8.5.1, 3.8.6.1, 3.8.7.1, 3.8.8.1, 3.8.9.1, 3.8.10.1, 3.8.11.1; tables I through XI).
e. Whether material shall be corrosion-resistant steel, galvanized steel, tin coated bronze or jacketed aramid fibre (see 3.2 and 3.7 and 3.8).
f. Type I grips. Specify eye configuration, closed or split mesh and lace method as applicable (see 3.8.1 and 3.8.2).
g. Type II grips. Specify ring support or hook ends, single or double weave, closed or split mesh and lace method and conduit size (see 3.8.3).
h. Type III grips, classes 1 through 4 and 6. Specify eye configuration and single or multiweave as applicable (see 3.8.4 through 3.8.7 and 3.8.9).
i. Type III, class 5 grips. Specify closed or split mesh and single or double eye (see 3.8.8).
j. Type IV grips. Specify flexible eye or ring, closed or split mesh, single or double weave and lace method as applicable (see 3.8.10 and 3.8.11).
k. Levels of preservation, packaging and marking required (see 5.1).
l. Military part number (see 6.4).
m. Whether first article is required (see 3.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the

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item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4.2. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Part or identifying number (PIN). The PIN to be used for cable grips acquired to this specification shall consist of the letter M, the basic number of this specification, and dash numbers compiled from characteristic codes, as shown in table XIV.

6.5 Component descriptions and definitions.

6.5.1 Body. The tubular, open-braid (mesh) section of the grip.

6.5.2 Braided stem. A node (area) at which the strand from the body (mesh) is woven into a prescribed braid for equalizing the loading and providing for the formation of the eye.

6.5.3 Closed. The closed grip is woven with no opening along the longitudinal axis of the body (mesh). It is used when the end of the object is available for assembly (see figure 1A).

6.5.4 Conduit riser grips. Conduit riser grips are used as a method of support for wire and cable, in ducts, conduit and similar applications (see figure 2).

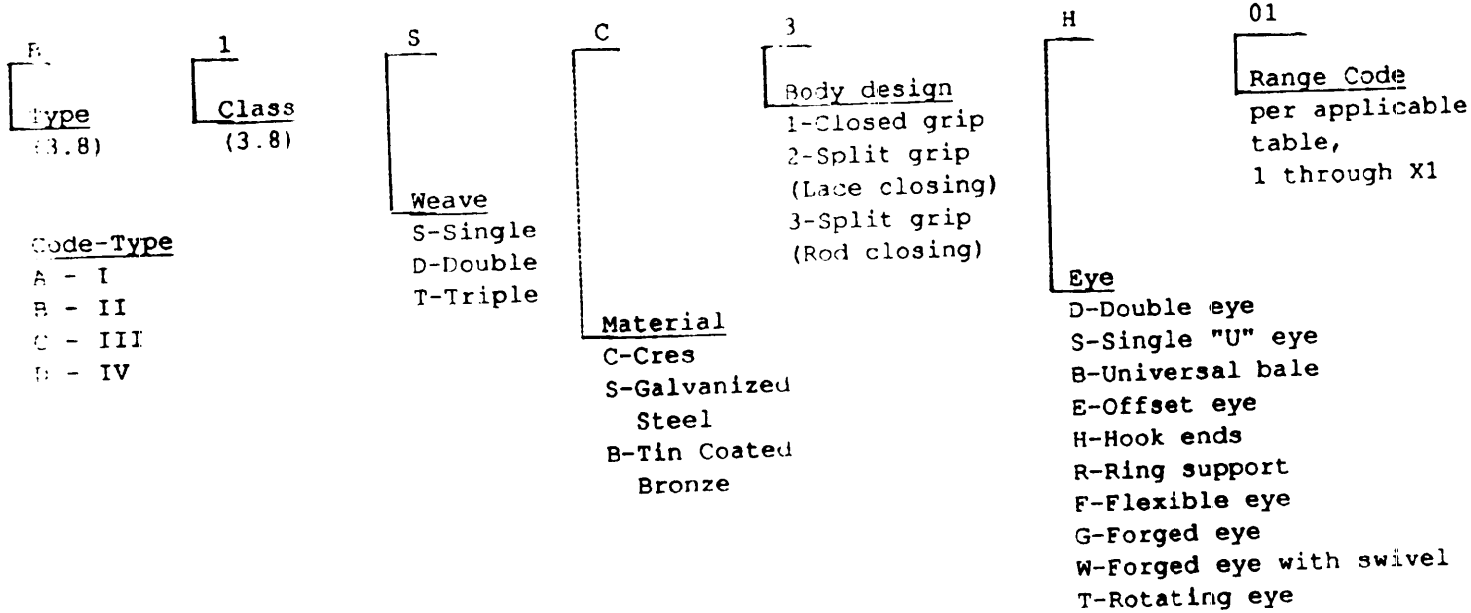
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Table XIV. Part or identifying numbe: (PIN) formulation.

EXAMPLE:

M16567

Mil Spec



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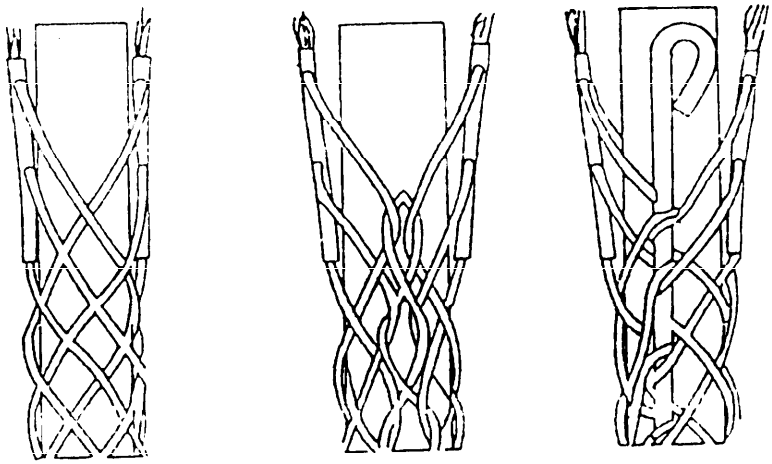


Figure 1 Grip, closed; split (lace closing) and split (rod closing).

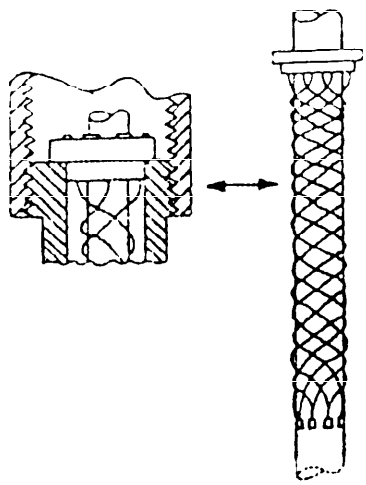


Figure 2. Conduit riser grip.

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6.5.5 End pull flexible eye. End pull flexible eye grips shall be as shown in figure 3.

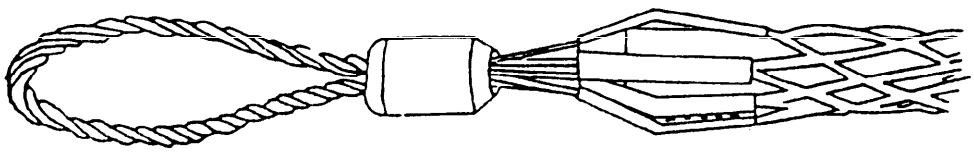


Figure 3. Flexible eye.

6.5.6 End pull rotating forged eye. End pull rotating forged eye grips shall be as shown in figure 4.

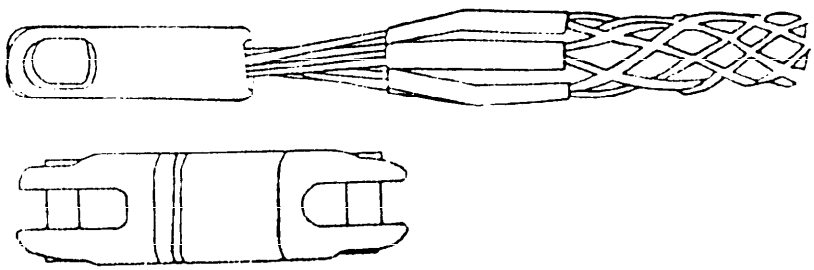


Figure 4. Forged eye with swivel (rotating).

6.5.7 Eye. The member of the grip which joins the grip to the supporting or pulling device.

6.5.8 Node. The intermediate member or area joining together the body and the eye.

6.5.9 Pulling grips. Pulling grips are used to pull cable, rope, or wire rope.

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6.5.10 Shoulder. A node (member) made of tubular material which provides for equalization of loading and protecting the strands from abrasion.

6.5.11 Slack pull (offset flexible eye). Slack pull (offset flexible eye) grips used for pulling slack, positioning and removing cable shall be as shown in figure 5.

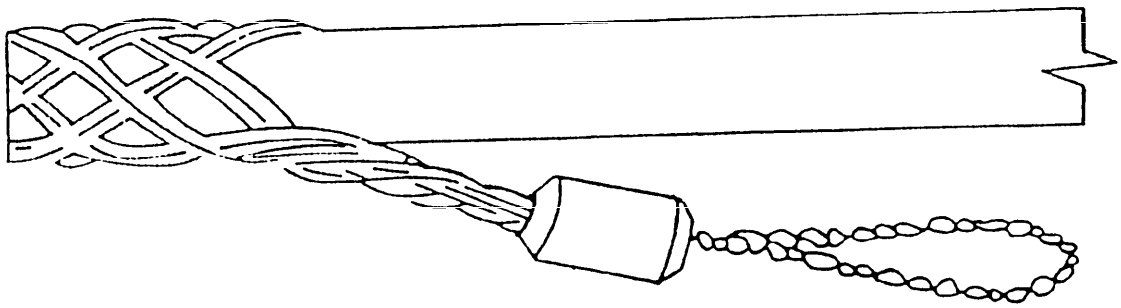


Figure 5. Slack pull, offset flexible eye.

6.5.12 Slack pull (ring type). Slack pulls (ring type) used for pulling slack, positioning and removing cable shall be as shown in figure 6.

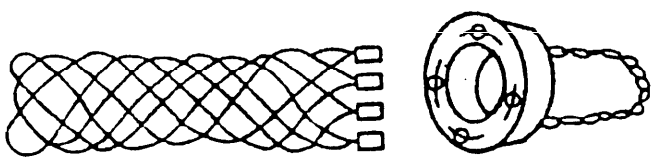


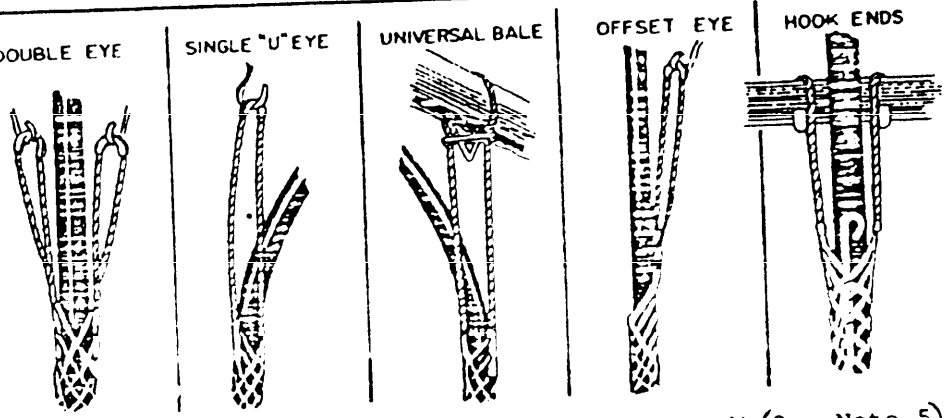
Figure 6. Slack pull, ring type.

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6.5.13 Split. The split grip is woven with a longitudinal separation in the body (mesh). It is sued when the end of the object is not available and can be opened and mounted (see figures 1B and 1C).

6.5.14 Strand equalizers. A node (member) made in tubular form to space the strand from the body (mesh) for the purpose of equalizing loading and providing for the formation of the eye.

6.5.15 Support riser grips. Support riser grips absorb flexure and variation and support cable weight (see figure 7).



NOTES:
1. Double eye is used where cable extends straight through grip without bending and where a double support is required.
2. Single eye is used where cable is vertical, where cable bends from vertical to horizontal, and where single support is required.
3. Universal bale is used where the support member is a pipe, beam, closed eye or continuous structural element.
4. Offset eye is used where a single offset support is required.
5. Hook ends type is used where cable extends straight through the grip and the support attachment is a steel plate.

Figure 7. Support riser grips.

6.5.16 Weave. The manner in which the body (mesh) of the grip is constructed. Single weave is one strand woven into a body. Double weave is two parallel, adjacent strands woven into a body. Triple weave is three parallel, adjacent strands woven into a body.

6.6 Subcontracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 of this specification do not apply when material and parts are procured by the supplier for incorporation into the equipment and thereby lose their separate identity when the equipment is shipped.

6.7 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

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6.7 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.8 Subject term (keyword) listing.

Cable ties
Handles
Holdfasts
Pulling devices
Towing

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

User Activity:
Army - CI

Preparing Activity:
Army - MI

Project No. 5120 - AC96

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